

Cross-Modal Embedding for Unified Memory Search

Abstract: We present RecallForge, a multimodal memory search system that embeds text, images, and video into a shared 2048-dimensional vector space using vision-language models. Our hybrid retrieval pipeline combines BM25 keyword search with dense vector similarity via reciprocal rank fusion, achieving 85% R@1 on text and 40% on cross-modal.

1. Introduction

Personal knowledge management systems have traditionally been limited to text search. Vision-language models like Qwen3-VL now enable unified semantic spaces for all content.

2. Architecture

Three-tier stack: embedder (Qwen3-VL-Embedding-2B), reranker (Qwen3-VL-Reranker-2B), captioner (Qwen3.5-0.8B). All run at 4-bit quantization on Apple Silicon via MLX.

3. Results

text_to_text: 85% R@1 (vector), 90% with reranker

text_to_image: 16.7% R@1 (vector), 40% with reranker

mixed_modal: 95% R@1 (vector), 100% with reranker